5.1 EIGENVECTORS AND EIGENVALUES

EXAMPLE 1 Let $A = \begin{bmatrix} 3 & -2 \\ 1 & 0 \end{bmatrix}$, $\mathbf{u} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$, and $\mathbf{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. The images of $\mathbf{u}$ and $\mathbf{v}$ under multiplication by A are shown in Figure 1. In fact, $A\mathbf{v}$ is just $2\mathbf{v}$. So A only "stretches," or dilates, $\mathbf{v}$. ■

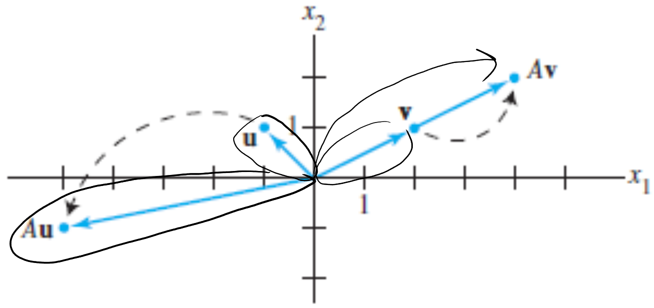

FIGURE 1 Effects of multiplication by A.

$T(u) = A\cdot u = \begin{bmatrix} -5 \\ -1 \end{bmatrix}$

$T(v) = A\cdot v = \begin{bmatrix} 4 \\ 2 \end{bmatrix} = 2\cdot\begin{bmatrix} 2 \\ 1 \end{bmatrix} =$

$= 2\cdot u$

$\boxed{Av = 2\cdot u}$

DEFINITION An **eigenvector** of an $n \times n$ matrix A is a nonzero vector $\mathbf{x}$ such that $A\mathbf{x} = \lambda\mathbf{x}$ for some scalar λ. A scalar λ is called an **eigenvalue** of A if there is a nontrivial solution $\mathbf{x}$ of $A\mathbf{x} = \lambda\mathbf{x}$; such an $\mathbf{x}$ is called an *eigenvector corresponding to* λ.[1]

λ – e/value, x – e/vector

It is easy to determine if a given vector is an eigenvector of a matrix. It is also easy to decide if a specified scalar is an eigenvalue.

EXAMPLE 2 Let $A = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$, $\mathbf{u} = \begin{bmatrix} 6 \\ -5 \end{bmatrix}$, and $\mathbf{v} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$. Are $\mathbf{u}$ and $\mathbf{v}$ eigenvectors of A?

$$A\cdot u = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}\cdot\begin{bmatrix} 6 \\ -5 \end{bmatrix} = \begin{bmatrix} -24 \\ 20 \end{bmatrix} = -4\cdot\begin{bmatrix} 6 \\ -5 \end{bmatrix} = -4\cdot u$$

$\begin{bmatrix} 12 \\ -10 \end{bmatrix}$ also e/vector check

Yes, $\bar{u}$ is an e/vector of A, cor. e/value is -4

$x = \begin{bmatrix} 6 \\ -5 \end{bmatrix}$

$\lambda = -4$

$$A v = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}\cdot\begin{bmatrix} 3 \\ -2 \end{bmatrix} = \begin{bmatrix} -9 \\ 11 \end{bmatrix} \neq \underset{-3}{\lambda}\cdot\begin{bmatrix} 3 \\ -2 \end{bmatrix}$$

No, v is not, no e/value.

EXAMPLE 3 Show that 7 is an eigenvalue of matrix A in Example 2, and find the corresponding eigenvectors.

$$\begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \lambda \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \underbrace{\lambda \cdot I \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}$$

$$A \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - 7 \cdot I \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$(A - \lambda I) \cdot \bar{x} = 0$$

must have non trivial sol.

$$(A - 7I) \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$ - system, homog

matrix 7

$$\begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix} - \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix} \Rightarrow \left[\begin{array}{cc|c} -6 & 6 & 0 \\ 5 & -5 & 0 \end{array}\right] \xrightarrow{RREF} \left[\begin{array}{cc|c} 1 & -1 & 0 \\ 1 & -1 & 0 \end{array}\right] \to \left[\begin{array}{cc|c} 1 & -1 & 0 \\ 0 & 0 & 0 \end{array}\right]$$

$$\{x_1 - x_2 = 0 \Rightarrow [x_1 = x_2] \Rightarrow e/v = \begin{bmatrix} 5 \\ 5 \end{bmatrix} \text{ or } \begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{ etc}$$

Warning: Although row reduction was used in Example 3 to find eigen*vectors*, it cannot be used to find eigen*values*. An echelon form of a matrix A usually does *not* display the eigenvalues of A.

The equivalence of equations (1) and (2) obviously holds for any λ in place of $\lambda = 7$. Thus λ is an eigenvalue of an $n \times n$ matrix A if and only if the equation

$$(A - \lambda I)\mathbf{x} = \mathbf{0} \quad \text{iff } |A - \lambda I| = 0 \qquad (3)$$

has a nontrivial solution. The set of *all* solutions of (3) is just the null space of the matrix $A - \lambda I$. So this set is a *subspace* of $\mathbb{R}^n$ and is called the **eigenspace** of A corresponding to λ. The eigenspace consists of the zero vector and all the eigenvectors corresponding to λ.

each e/value has inf. many e/vectors completing e/space cor to λ

Example 3 shows that for matrix A in Example 2, the eigenspace corresponding to $\lambda = 7$ consists of *all* multiples of (1, 1), which is the line through (1, 1) and the origin. From Example 2, you can check that the eigenspace corresponding to $\lambda = -4$ is the line through $(6, -5)$. These eigenspaces are shown in Figure 2, along with eigenvectors (1, 1) and $(3/2, -5/4)$ and the geometric action of the transformation $\mathbf{x} \mapsto A\mathbf{x}$ on each eigenspace.

EXAMPLE 4 Let $A = \begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix}$. An eigenvalue of A is 2. Find a basis for the corresponding eigenspace.

$$\left[\underbrace{\begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix}}_{A} - \underbrace{\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}}_{2I}\right] \cdot \bar{x} = 0$$

$$\begin{bmatrix} 2 & -1 & 6 \\ 2 & -1 & 6 \\ 2 & -1 & 6 \end{bmatrix} \begin{matrix} R_1/2 \\ R_2 - R_1 \\ R_3 - R_1 \end{matrix} \quad \left[\begin{array}{ccc|c} \overset{\text{basic}}{1} & \overset{\text{free}}{-1/2} & \overset{\text{free}}{3} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \Rightarrow x_1 - \tfrac{1}{2}x_2 + 3x_3 = 0$$

$$x_1 = \tfrac{1}{2}x_2 - 3x_3 \iff x_2 = s, \quad x_3 = t$$

$$\bar{x} = s \cdot \begin{bmatrix} 1/2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} \Rightarrow \underbrace{\bar{x} = \text{span}}_{\text{e/space of } \lambda = 2} \left\{ \overset{V_1}{\begin{bmatrix} 1/2 \\ 1 \\ 0 \end{bmatrix}}, \overset{V_2}{\begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}} \right\}$$

Basis $= \{V_1, V_2\}$ of e/space

THEOREM 1 The eigenvalues of a triangular matrix are the entries on its main diagonal.

PROOF For simplicity, consider the 3×3 case. If A is upper triangular, then $A - \lambda I$ has the form

$$A - \lambda I = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$= \begin{bmatrix} a_{11} - \lambda & a_{12} & a_{13} \\ 0 & a_{22} - \lambda & a_{23} \\ 0 & 0 & a_{33} - \lambda \end{bmatrix} \cdot X = 0$$

$$a_{11} - \lambda = 0 \qquad a_{22} - \lambda = 0 \qquad a_{33} - \lambda = 0$$
$$\lambda = a_{11}$$

The scalar λ is an eigenvalue of A if and only if the equation $(A - \lambda I)\mathbf{x} = \mathbf{0}$ has a nontrivial solution, that is, if and only if the equation has a free variable. Because of the zero entries in $A - \lambda I$, it is easy to see that $(A - \lambda I)\mathbf{x} = \mathbf{0}$ has a free variable if and only if at least one of the entries on the diagonal of $A - \lambda I$ is zero. This happens if and only if λ equals one of the entries a_{11}, a_{22}, a_{33} in A. For the case in which A is lower triangular, see Exercise 28. ■

If $\mathbf{v}_1, \ldots, \mathbf{v}_r$ are eigenvectors that correspond to distinct eigenvalues $\lambda_1, \ldots, \lambda_r$ of an $n \times n$ matrix A, then the set $\{\mathbf{v}_1, \ldots, \mathbf{v}_r\}$ is linearly independent.

Proof: Let $\{v_1 \ldots v_r\}$ be a $\boxed{\text{lin dep}}$ false

$\exists$ $(*)\ \bar{v}_{p+1} = c_1\bar{v}_1 + c_2\bar{v}_2 + \ldots + c_p\bar{v}_p$ (others $\bar{v}_{p+2} \ldots \bar{v}_r$ may be zero vectors)

$A \cdot \bar{v}_{p+1} = A \cdot c_1\bar{v}_1 + A \cdot c_2\bar{v}_2 + \ldots + A \cdot c_p \cdot \bar{v}_p$ ($v_1 \ldots v_p$ - lin indep)

$\lambda_{p+1} \cdot \bar{v}_{p+1} = c_1 \cdot \lambda_1\bar{v}_1 + c_2 \cdot \lambda_2\bar{v}_2 + \ldots + c_p \cdot \lambda_p \cdot \bar{v}_p$

$(*) \cdot \lambda_{p+1} \Rightarrow \lambda_{p+1} \cdot \bar{v}_{p+1} = c_1 \cdot \lambda_{p+1} \cdot \bar{v}_1 + \ldots + c_p \cdot \lambda_{p+1} \cdot \bar{v}_p$ $-$

$\bar{0} = c_1 \cdot (\lambda_1 - \lambda_{p+1}) \cdot \bar{v}_1 + c_2(\lambda_2 - \lambda_{p+1})\bar{v}_2 + \ldots + c_p(\lambda_p - \lambda_{p+1}) \cdot \bar{v}_p$

$c_1 \neq 0 \Rightarrow \lambda_1 - \lambda_{p+1} = 0$

$c_i \neq 0 \Rightarrow \boxed{\lambda_i - \lambda_{p+1} = 0} \Rightarrow \lambda_i = \lambda_{p+1}$ ↯

$\Rightarrow v_1 \ldots v_p$ - lin indep

5.2 THE CHARACTERISTIC EQUATION

Useful information about the eigenvalues of a square matrix A is encoded in a special scalar equation called the characteristic equation of A. A simple example will lead to the general case.

EXAMPLE 1 Find the eigenvalues of $A = \begin{bmatrix} 2 & 3 \\ 3 & -6 \end{bmatrix}$.

$|A - \lambda I| = 0 \rightarrow$ c/e

$\det \left[\begin{pmatrix} 2 & 3 \\ 3 & -6 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right] = 0$

$\begin{vmatrix} 2-\lambda & 3 \\ 3 & -6-\lambda \end{vmatrix} = 0$ $\quad (2-\lambda)\cdot(-6-\lambda) - 3\cdot 3 = 0$

$-12 - \underline{2\lambda} + \underline{6\lambda} + \lambda^2 - 9 = 0$

$\lambda^2 + 4\lambda - 21 = 0$

$\downarrow$ char. equat

$\lambda_1 = -7, \lambda_2 = 3$

$\downarrow$

e/values are

$-7, 3$

The Invertible Matrix Theorem (continued)

Let A be an $n \times n$ matrix. Then A is invertible if and only if: $\exists A^{-1}$ iff $|A| \neq 0$

s. The number 0 is *not* an eigenvalue of A.

t. The determinant of ~~A is *not*~~ zero.

The Characteristic Equation

Theorem 3(a) shows how to determine when a matrix of the form $A - \lambda I$ is ~~not~~ invertible. The scalar equation $\det(A - \lambda I) = 0$ is called the **characteristic equation** of A, and the argument in Example 1 justifies the following fact.

> A scalar λ is an eigenvalue of an $n \times n$ matrix A if and only if λ satisfies the characteristic equation
>
> $$\det(A - \lambda I) = 0$$

EXAMPLE 3 Find the characteristic equation of

$$A = \begin{bmatrix} 5 & -2 & 6 & -1 \\ 0 & 3 & -8 & 0 \\ 0 & 0 & 5 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

upper triangular

SOLUTION Form $A - \lambda I$, and use ~~Theorem~~ 3(d):

$$\det(A - \lambda I) = \det \begin{bmatrix} 5-\lambda & -2 & 6 & -1 \\ 0 & 3-\lambda & -8 & 0 \\ 0 & 0 & 5-\lambda & 4 \\ 0 & 0 & 0 & 1-\lambda \end{bmatrix}$$

$$= \underbrace{(5-\lambda)(3-\lambda)(5-\lambda)(1-\lambda)}$$

$\lambda_{1,2} = 5$

$\lambda_3 = 1$

$\lambda_4 = 3$

$$A = \begin{bmatrix} 4-\lambda & 0 & 1 \\ -2 & 1-\lambda & 0 \\ -2 & 0 & 1-\lambda \end{bmatrix} \rightarrow \begin{vmatrix} 4-\lambda & 0 & 1 \\ -2 & 1-\lambda & 0 \\ -2 & 0 & 1-\lambda \end{vmatrix} = 0$$

$$(-1)^{2+2}(1-\lambda)\cdot\begin{vmatrix} 4-\lambda & 1 \\ -2 & 1-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)\left((4-\lambda)(1-\lambda) - 1\cdot(-2)\right) = 0$$

$$(1-\lambda)(\lambda^2 - 5\lambda + 6) = 0$$

$$\lambda_1 = 1$$

$$\begin{bmatrix} 3 & 0 & 1 \\ -2 & 0 & 0 \\ -2 & 0 & 0 \end{bmatrix} \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 1/3 & 0 \\ 0 & 0 & 1/3 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\begin{cases} x_1 + \frac{1}{3}x_3 = 0 \\ \frac{1}{3}x_3 = 0 \end{cases}$$

$x_3 = 0$ $\quad \bar{X} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

$x_1 = 0$

$x_2 =$ free

$$e/space = span\left\{\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}\right\} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ \sqrt{5} \\ 0 \end{bmatrix}, \ldots = c\cdot\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, c \in \mathbb{R}$$

$$\lambda_2 = 2$$

$$\begin{bmatrix} 2 & 0 & 1 \\ -2 & -1 & 0 \\ -2 & 0 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1/2 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{cases} x_1 + \frac{1}{2}x_3 = 0 \\ x_2 - x_3 = 0 \end{cases} \quad x_3 \rightarrow s$$

$$\begin{cases} x_1 = -\frac{1}{2}s \\ x_2 = s \\ x_3 = s \end{cases} \quad \bar{X} = \begin{bmatrix} -1/2 \\ 1 \\ 1 \end{bmatrix}$$

$$e/space = span\left\{\begin{bmatrix} -1/2 \\ 1 \\ 1 \end{bmatrix}\right\} = \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix}, \begin{bmatrix} -2,5 \\ 5 \\ 5 \end{bmatrix}, \ldots = c\cdot\begin{bmatrix} -1/2 \\ 1 \\ 1 \end{bmatrix}, c \in \mathbb{R}$$

$$\lambda_3 = 3$$

$$\begin{bmatrix} 1 & 0 & 1 \\ -2 & -2 & 0 \\ -2 & 0 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & -2 & 2 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{cases} x_1 + x_3 = 0 \\ x_2 - x_3 = 0 \end{cases}$$

$$\begin{cases} x_1 = -t \\ x_2 = t \\ x_3 = t \end{cases} \quad \bar{X} = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

$$e/space = span\left\{\begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}\right\} = c\cdot\begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}, c \in \mathbb{R}$$

$$A = \begin{bmatrix} 4-\lambda & 2 & 3 \\ -1 & 1-\lambda & -3 \\ 2 & 4 & 9-\lambda \end{bmatrix} \to \begin{vmatrix} 4-\lambda & 2 & 3 \\ -1 & 1-\lambda & -3 \\ 2 & 4 & 9-\lambda \end{vmatrix} = 0$$

$$(-1)^{1+1}\cdot(4-\lambda)\begin{vmatrix} 1-\lambda & -3 \\ 4 & 9-\lambda \end{vmatrix} + (-1)^{1+2}\cdot 2\cdot\begin{vmatrix} -1 & -3 \\ 2 & 9-\lambda \end{vmatrix} + (-1)^{1+3}\cdot 3\cdot\begin{vmatrix} -1 & 1-\lambda \\ 2 & 4 \end{vmatrix} = 0$$

$$(4-\lambda)\Big((1-\lambda)(9-\lambda)+12\Big) - 2\Big(-1(9-\lambda)+6\Big) + 3\cdot\Big(-1\cdot 4 - 2\cdot(1-\lambda)\Big) = 0$$

$$\begin{array}{r} 12 \\ 9 \\ \hline 108 \end{array}$$

$$(4-\lambda)(9-\lambda-9\lambda+12+\lambda^2) - 2(-9+\lambda+6) + 3(-4-2+2\lambda) = 0$$

$$(4-\lambda)(\lambda^2-10\lambda+108) - 2(\lambda-3) + 3(2\lambda-6) = 0$$

$$4\lambda^2 - 40\lambda + 432 - \lambda^3 + 10\lambda^2 - 108\lambda + 4\lambda - 12 = 0$$

$$\lambda^3 - 14\lambda^2 + 144\lambda - 420 = 0$$

$$\lambda_1 \approx 4.05, \quad \lambda_{2,3} - \mathbb{C}$$